

Report: Gossip & Push-Sum Convergence vs Network Size

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- Actor-model simulation in Gleam (one actor per node + coordinator).
- Convergence time printed from monotonic clock (native ticks, converted to ms ticks/1e6).

Key Observations (from these runs):

- Full topology is fastest;
- line topology is slowest due to long diameter.
- Imperfect 3D is consistently faster than plain 3D (shortcut neighbor).
- Gossip scales well to large n (50k 100k+) depending on topology;
- convergence increases roughly with path length / node degree trade-offs.
- Push-sum is slower and more variable; stability relies on epsilon and continued relaying.
- For very small or very large graphs, a slightly looser epsilon (e.g., $1e-8$) improves results.

Notes:

- Exact ms depend on the VM/machine.
- These plots use ms ticks / 1,000,000. Consider log-scale y-axis for readability over large ranges.

Gossip: Convergence Time vs Network Size



